# TECHNICAL REPORT FOR THE ICASSP 2024 CADENZA GRAND CHALLENGE

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## ABSTRACT

In this report, we briefly describe the system we use to participate the Cadenza challenge. Our system builds on top of hybrid Demucs (hdemucs) [1], one of the provided baselines. The main skeleton of our model is a Spec-UNet, with complex-as-channel (CAC) input. Further, we replace the original complex ratio mask with a deep filter from [2]. Our system demonstrates an increment in HAAQI metric [3] from the baseline hdemucs system when tested against the validation set.

***Index Terms*—** Hearing Aids, Music Enhancement

## 1. INTRODUCTION

ICASSP 2024 Cadenza Challenge focuses on how to enhance music perception for those with hearing aids. While general audio enhancement and speech enhancement are well researched topics and advance rapidly with the rise of deep learning techniques, enhancement techniques that particularly target people with hearing disabilities receive less attention.

The challenge attempts to address this situation via a separation-remix pipeline. As shown in Figure 1, we start with the mixtures from MUSDB18 dataset. Applying HRTFs to them gives us the mixtures at the hearing aids' microphone (mixtures-at-mic). The HRTFs application is considered a (and the only) preprocessing step, and we will start with mixtures-at-mic during test time.

Now given the pretrained baseline systems, such as hdemucs and open-unmix, we feed the mixtures-at-mic of a particular scene in and obtain the separated stems. We then apply gains associated with that scene to each stem and remix them. Finally, we apply the listener characteristics on the remix via NALR, which essentially reduces to a convolution operation in the waveform domain. Note that listeners are not associated with scenes and they can be chosen arbitrarily.

The NALRed remix is then compared against the ground truth remix, which consists of processing individual ground truth stems from MUSDB18 in an identical manner as the predicted remix. The comparison metric used is HAAQI, which, comparing to the conventional SDR/SI-SDR, focuses on envelope and temporal fine structure and is shown to better predict the perceptual quality [3].

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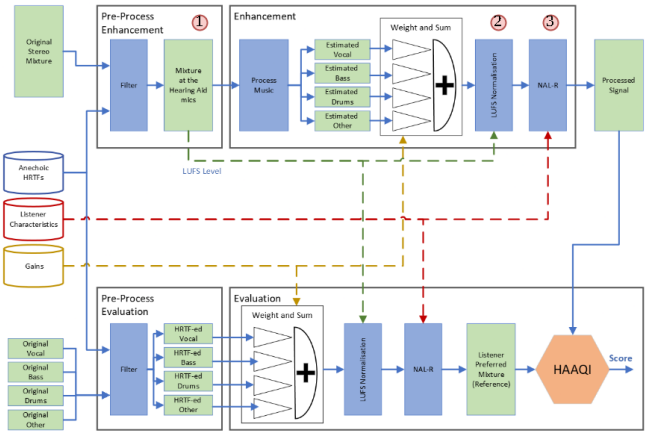


**Fig. 1**. Cadenza Challenge's pipeline. The circled number 1, 2 and 3 indicate components of the input to our designed model, whose output then replaces the role of 3 and compares against the ground truth remix

To further improve this pipeline, we make two observations. First, instead of trying to design a better separation network that outcompetes the baselines, one can attempt to simply attach a extra fine-tuning network right after the baseline networks to correct the distribution-shift caused by applying HRTFs. Further, one can adjust the location of the fine-tuning network to the end of the pipeline, that is, after applying NALRs, to take into account information about the gains and NALRs

Also, given that HAAQI focus on temporal fine structures, rather than outputing a complex ratio mask at the end, we borrow the idea of deep filter from [2]. As shown in [2], deep filter concentrate its estimation effort within a small temporal window and thus can handle temporal fine structure better than the cRM.

## 2. THEORY

### 2.1. Model Input and Architecture

Our model takes in a concatenation of at-mic-mixture, remix after the separation baseline, and NALRed remix as input. Before the concatenation, we perform STFT to each of the mixtures. Hence, at the entrance of the network, our data shape becomes $\mathbb{C}^{B\times 3\times T\times F} \rightarrow \mathbb{R}^{B\times 6\times T\times F}$, where $B$ is the

batch size, $T$ is the sampled time steps and $F$ is the frequency bins and the arrow indicates complex-as-channel operation.

Now the data pack simply passes through a convolutional unet-like structure. Each convolutional block in the encoder will double the channel size, while average-pooling the time and frequency axis each by a facotr of 2. The exception is the first block, where we direectly elevate the number of channels to 32. With a total of 7 such blocks, by the time the data reaches the bottleneck, the shape become $\mathbb{R}^{B\times 2048\times T/2^{L-1}\times F/2^{L-1}}$, where $L$ is the number of convolution blocks in the encoder. Similarly, each convolution block in the decoder inverts the operation of the corresponding encoder block via transpose convolution. Once we return to 32 channels, we can either further shrink down the channels to produce a complex ratio mask, or map it to a deep filter, as explained in the following section. The final output has shape $\mathbb{R}^{B\times 4\times T\times F} \rightarrow \mathbb{C}^{B\times 2\times T\times F}$.

Finally, inverse STFT is performed and its output is trained against ground-truth remix using mean absolute error loss.

### 2.2. Deep Filter

As explained in [2], deep filter is a temporal filter of shape $\mathbb{C}^{N\times F}$ that acts on a spectrogram of the shape $\mathbb{C}^{T\times F}$. The detailed operation consists of unfolding the spectrogram along the time axis with kernel length $N$ so that the shape becomes $\mathbb{C}^{T\times N\times F}$. Then we simply perform the dot product between the deep filter and the spectrogram along the $N$ axis to obtain $\mathbb{C}^{T\times F}$. Notice the lookahead created by the deepfilter depends on how we unfold the spectrogram. If the unfolding kernel centers at each time point, then lookahead will be $N//2$. Similarly, if the kernel has its right end at each time point, then the filters become causal. After the deep filter, we perform the inverse STFT and the computation afterwards remains identical to the complex ratio mask case.

For the sake of completeness, we also attempted to include the ERB encoders of the deepfilternet into our unet. But we do not see significant increment in terms of the validation metrics.

## 3. EXPERIMENTS

### 3.1. Datasets and Experimental setup

We use the provided MUSDB18 training subset as the training set. No augmentation is performed and extra datasets, including MedleyDB, Bach10, FMA, are not used. For hyperparameters of Spec-U-Net, we use a window size and FFT size of 2048, hop size of 441. Since all of our inputs use 44100 Hz as the sampling rate, our output remains consistent to that. The number of convolutional blocks in both the encoder and the decoder is $L = 7$. For the deep filter, we set the unfolding kernel length to $N = 5$. Higher kernel lengths produce similar performance.

For the general training hyperparameters, the batch size is 36. At training time, we randomly cut 2-second clips from the pool of songs. The initial learning rate is set to 1e-3 with no decaying multiplier. We implemented all methods in Pytorch and trained on a NVIDIA A6000 GPU with a maximum epoch of 500 and it takes less than one day to complete one pass of training. Since HAAQI's computation is slow, we use the mean SDR as our evaluation metric during the validation.

| | hdemucs | Our model (cRM) | Our model (DF) |
|---|---|---|---|
| mean SDR | 11.047 | 11.189 | 11.221 |
| left HAAQI | 0.6690 | 0.6759 | 0.6786 |
| right HAAQI | 0.6665 | 0.6705 | 0.6798 |
| overall HAAQI | 0.6677 | 0.6732 | 0.6792 |

**Table 1**. Evaluation Results. cRM refers to complex ratio mask and DF refers to deep filter at the output of our model

### 3.2. Results and Discussion

We list the metrics in Table 1. As we can see, both SDRs and HAAQI metric marginally increases with the complex ratio mask version of our model. The metrics further increase a bit when we replace the cRM with deep filters. The reason that the metrics do not significantly increase is likely due to the fact that the model is having trouble generalizing to the listeners that it has not seen before. For example, if instead of using a separate validation set, we split the current training set into a training subset and a validation subset, then SDR increment can boost up to 0.5 dB as the listeners in this situation will be known to the model during the validation.

## 4. CONCLUSION

In conclusion, we design a system that builds upon hybrid demucs to enhance its performance. Results of SDR and HAAQI metrics, while confirming the positive effect of our design in improving perception, also indicate the difficulty of generalization beyond seen listeners. We will investigate this issue in the future work.

## 5. REFERENCES


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